2026

HEALTH ANALYSIS REPORT

ADVANCE POWER-BI EXAMINATION

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Part A: Data Acquisition and Understanding

1. Domain and Business Problem

Domain: Healthcare Analytics

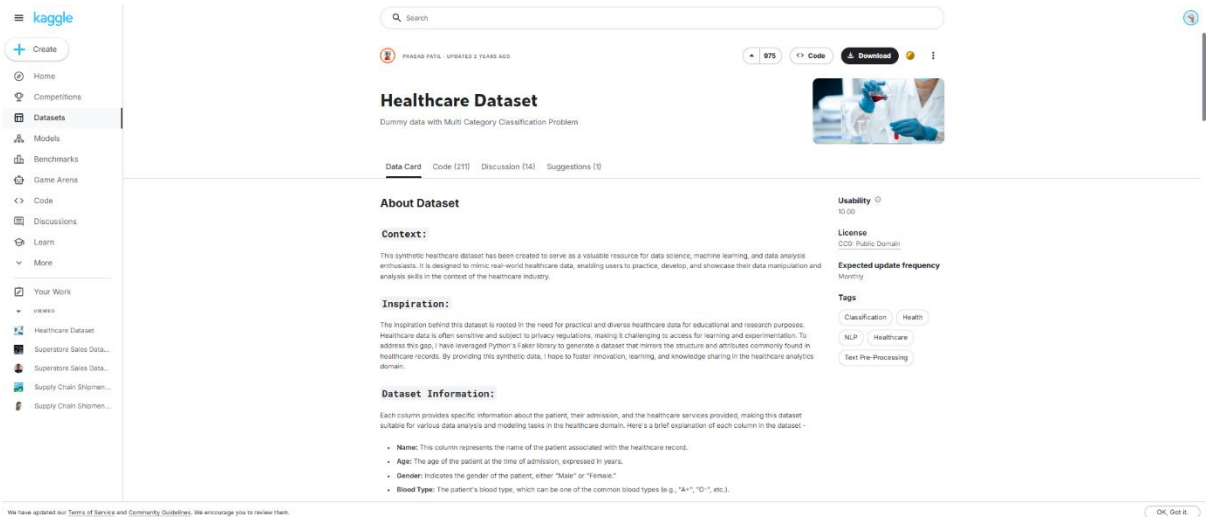
Business Problem: Healthcare institutions generate vast amounts of patient data daily, yet many administrators lack a unified view of admissions, clinical outcomes, and financial performance. This project addresses the following core business question:

"How can hospital administrators monitor patient admission patterns, understand the financial burden of different medical conditions, track clinical outcomes, and identify high-risk patient segments so that they are able to improve resource allocation, reduce costs, and enhance quality of care?"

Key decision-making questions this solution answers:

- Which medical conditions drive the highest billing costs?
- Are emergency admissions significantly more expensive than elective ones?
- Which hospitals and doctors are handling the highest patient loads?
- Is there a seasonal pattern in patient admissions across the year?
- What proportion of patients have abnormal test results, and how does that correlate with length of stay?

2. Data Source



The dataset was created to simulate real-world hospital records while protecting patient privacy. It mirrors the structure and attributes commonly found in actual healthcare records, making it suitable for educational and analytical purposes.

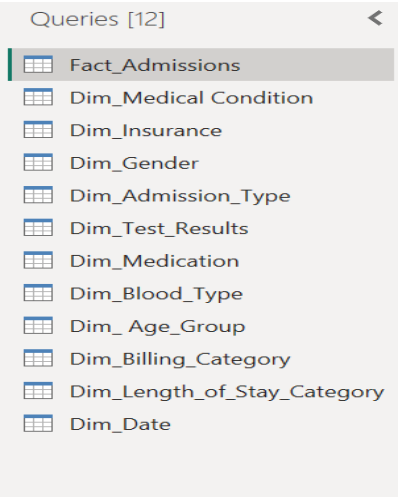
3. Key Tables and Fields

Dataset Information:

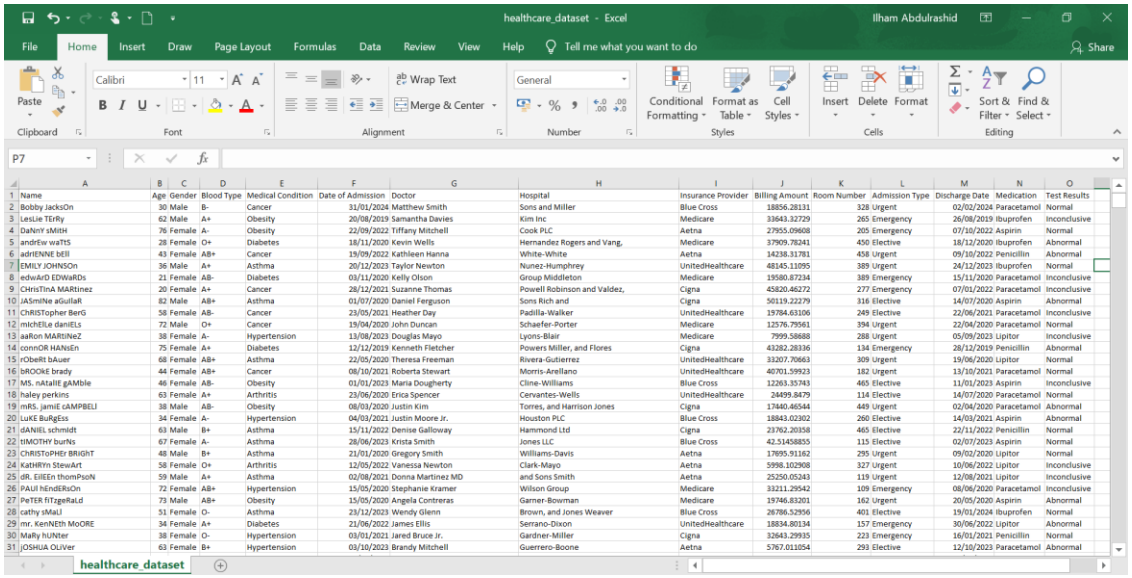
Each column provides specific information about the patient, their admission, and the healthcare services provided, making this dataset suitable for various data analysis and modeling tasks in the healthcare domain. Here's a brief explanation of each column in the dataset -

- **Name:** This column represents the name of the patient associated with the healthcare record.
- **Age:** The age of the patient at the time of admission, expressed in years.
- **Gender:** Indicates the gender of the patient, either "Male" or "Female."
- **Blood Type:** The patient's blood type, which can be one of the common blood types (e.g., "A+", "O-", etc.).
- **Medical Condition:** This column specifies the primary medical condition or diagnosis associated with the patient, such as "Diabetes," "Hypertension," "Asthma," and more.
- **Date of Admission:** The date on which the patient was admitted to the healthcare facility.
- **Doctor:** The name of the doctor responsible for the patient's care during their admission.
- **Hospital:** Identifies the healthcare facility or hospital where the patient was admitted.
- **Insurance Provider:** This column indicates the patient's insurance provider, which can be one of several options, including "Aetna," "Blue Cross," "Cigna," "UnitedHealthcare," and "Medicare."
- **Billing Amount:** The amount of money billed for the patient's healthcare services during their admission. This is expressed as a floating-point number.
- **Room Number:** The room number where the patient was accommodated during their admission.
- **Admission Type:** Specifies the type of admission, which can be "Emergency," "Elective," or "Urgent," reflecting the circumstances of the admission.
- **Discharge Date:** The date on which the patient was discharged from the healthcare facility, based on the admission date and a random number of days within a realistic range.
- **Medication:** Identifies a medication prescribed or administered to the patient during their admission. Examples include "Aspirin," "Ibuprofen," "Penicillin," "Paracetamol," and "Lipitor."
- **Test Results:** Describes the results of a medical test conducted during the patient's admission. Possible values include "Normal," "Abnormal," or "Inconclusive," indicating the outcome of the test.

Tables:



4. Raw Data



Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	Hospital	Insurance Provider	Billing Amount	Room Number	Admission Type	Discharge Date	Medication	Test Results
Bobby JacksOn	30	Male	B-	Cancer	31/01/2024	Matthew Smith	Sons and Miller	Blue Cross	18856.28131	328	Urgent	02/02/2024	Paracetamol	Normal
LesLie TErRy	62	Male	A+	Obesity	20/08/2019	Samantha Davies	Kim Inc	Medicare	33643.32729	265	Emergency	26/08/2019	Ibuprofen	Inconclusive
DaNnY sMitH	76	Female	A-	Obesity	22/09/2022	Tiffany Mitchell	Cook PLC	Aetna	27955.09608	205	Emergency	07/10/2022	Aspirin	Normal
andrEw waTtS	28	Female	O+	Diabetes	18/11/2020	Kevin Wells	Hernandez Rogers and Vang,	Medicare	37909.78241	450	Elective	18/12/2020	Ibuprofen	Abnormal
adrIENNE bEll	43	Female	AB+	Cancer	19/09/2022	Kathleen Hanna	White-White	Aetna	14238.31781	458	Urgent	09/10/2022	Penicillin	Abnormal
EMILY JOHNSOn	36	Male	A+	Asthma	20/12/2023	Taylor Newton	Nunez-Humphrey	UnitedHealthcare	48145.11095	389	Urgent	24/12/2023	Ibuprofen	Normal
edwArD EDWaRDs	21	Female	AB-	Diabetes	03/11/2020	Kelly Olson	Group Middleton	Medicare	19580.87234	389	Emergency	15/11/2020	Paracetamol	Inconclusive
CHrisTInA MARtInez	20	Female	A+	Cancer	28/12/2021	Suzanne Thomas	Powell Robinson and Valdez,	Cigna	45820.46272	277	Emergency	07/01/2022	Paracetamol	Inconclusive
JASmINe aGuIlaR	82	Male	AB+	Asthma	01/07/2020	Daniel Ferguson	Sons Rich and	Cigna	50119.22279	316	Elective	14/07/2020	Aspirin	Abnormal
ChRISTopher BerG	58	Female	AB-	Cancer	23/05/2021	Heather Day	Padilla-Walker	UnitedHealthcare	19784.63106	249	Elective	22/06/2021	Paracetamol	Inconclusive
mIchElLe danIELs	72	Male	O+	Cancer	19/04/2020	John Duncan	Schaefer-Porter	Medicare	12576.79561	394	Urgent	22/04/2020	Paracetamol	Normal
aaRon MARtINeZ	38	Female	A-	Hypertension	13/08/2023	Douglas Mayo	Lyons-Blair	Medicare	7999.58688	288	Urgent	05/09/2023	Lipitor	Inconclusive
connOR HANsEn	75	Female	A+	Diabetes	12/12/2019	Kenneth Fletcher	Powers Miller, and Flores	Cigna	43282.28336	134	Emergency	28/12/2019	Penicillin	Abnormal
rObeRt bAuer	68	Female	AB+	Asthma	22/05/2020	Theresa Freeman	Rivera-Gutierrez	UnitedHealthcare	33207.70663	309	Urgent	19/06/2020	Lipitor	Normal
bROOkE brady	44	Female	AB+	Cancer	08/10/2021	Roberta Stewart	Morris-Arellano	UnitedHealthcare	40701.59923	182	Urgent	13/10/2021	Paracetamol	Normal
MS. nAtaliE gAMble	46	Female	AB-	Obesity	01/01/2023	Maria Dougherty	Cline-Williams	Blue Cross	12263.35743	465	Elective	11/01/2023	Aspirin	Inconclusive
haley perkins	63	Female	A+	Arthritis	23/06/2020	Erica Spencer	Cervantes-Wells	UnitedHealthcare	24499.8479	114	Elective	14/07/2020	Paracetamol	Normal
mRS. jamiE cAMPBELl	38	Male	AB-	Obesity	08/03/2020	Justin Kim	Torres, and Harrison Jones	Cigna	17440.46544	449	Urgent	02/04/2020	Paracetamol	Abnormal
LuKE BuRgEss	34	Female	A-	Hypertension	04/03/2021	Justin Moore Jr.	Houston PLC	Blue Cross	18843.02302	260	Elective	14/03/2021	Aspirin	Abnormal
dANIEL schmIdt	63	Male	B+	Asthma	15/11/2022	Denise Galloway	Hammond Ltd	Cigna	23762.20358	465	Elective	22/11/2022	Penicillin	Normal
tIMOTHY burNs	67	Female	A-	Asthma	28/06/2023	Krista Smith	Jones LLC	Blue Cross	42.51458855	115	Elective	02/07/2023	Aspirin	Normal
ChRISToPHEr BRIGhT	48	Male	B+	Asthma	21/01/2020	Gregory Smith	Williams-Davis	Aetna	17695.91162	295	Urgent	09/02/2020	Lipitor	Normal
KatHRYn StewArt	58	Female	O+	Arthritis	12/05/2022	Vanessa Newton	Clark-Mayo	Aetna	5998.102908	327	Urgent	10/06/2022	Lipitor	Inconclusive
dR. EIlEEn thomPsoN	59	Male	A+	Asthma	02/08/2021	Donna Martinez MD	and Sons Smith	Aetna	25250.05243	119	Urgent	12/08/2021	Lipitor	Inconclusive
PAUl hENdERsOn	72	Female	AB+	Hypertension	15/05/2020	Stephanie Kramer	Wilson Group	Medicare	33211.29542	109	Emergency	08/06/2020	Paracetamol	Inconclusive
PeTER fiTzgeRaLd	73	Male	AB+	Obesity	15/05/2020	Angela Contreras	Garner-Bowman	Medicare	19746.83201	162	Urgent	20/05/2020	Aspirin	Abnormal
cathy sMaLl	51	Female	O-	Asthma	23/12/2023	Wendy Glenn	Brown, and Jones Weaver	Blue Cross	26786.52956	401	Elective	19/01/2024	Ibuprofen	Normal
mr. KenNEth MooRE	34	Female	A+	Diabetes	21/06/2022	James Ellis	Serrano-Dixon	UnitedHealthcare	18834.80134	157	Emergency	30/06/2022	Lipitor	Abnormal
MaRy hUNter	38	Female	O-	Hypertension	03/01/2021	Jared Bruce Jr.	Gardner-Miller	Cigna	32643.29935	223	Emergency	16/01/2021	Penicillin	Normal
JOSHUA OLIVer	63	Female	B+	Hypertension	03/10/2023	Brandy Mitchell	Guerrero-Boone	Aetna	5767.011054	293	Elective	12/10/2023	Paracetamol	Abnormal

5. Why this dataset is suitable for advanced Power BI Analysis

This dataset is suitable for advanced Power BI analysis because it has enough data and the right structure.

It contains **10,000 records**, which is enough to show clear trends and patterns.

It includes both **categorical data** (like gender, medical condition, and insurance provider) and **numerical data** (like age and billing amount), which allows for different types of analysis.

There are also **two date fields**, which make it possible to do time-based analysis like trends over time and length of stay.

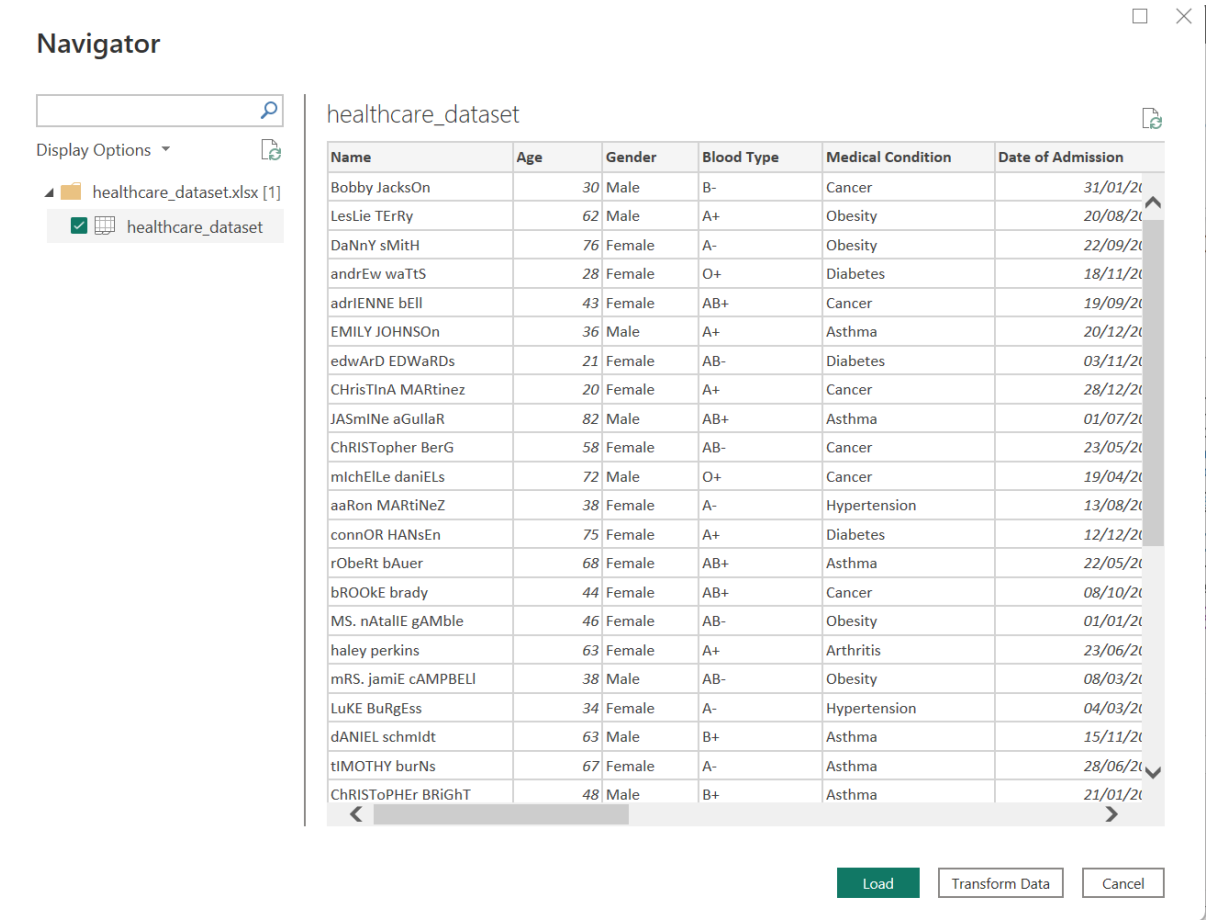
The dataset is well-structured, so it supports a **star schema**, which improves performance and makes analysis more efficient.

It also allows for creating useful **DAX measures** like total billing, averages, and percentages.

Finally, the dataset covers different areas like financial, clinical, and demographic data, making it useful for building multiple dashboards.

Part B: Data Cleaning and Transformation in Power Query

1. Data Import



Name	Age	Gender	Blood Type	Medical Condition	Date of Admission
Bobby JacksOn	30	Male	B-	Cancer	31/01/2
LesLie TErRy	62	Male	A+	Obesity	20/08/2
DaNnY sMitH	76	Female	A-	Obesity	22/09/2
andrEw waTtS	28	Female	O+	Diabetes	18/11/2
adrIENNE bEll	43	Female	AB+	Cancer	19/09/2
EMILY JOHNSOn	36	Male	A+	Asthma	20/12/2
edwArD EDWaRDs	21	Female	AB-	Diabetes	03/11/2
CHrisTInA MARtinez	20	Female	A+	Cancer	28/12/2
JASmINe aGullaR	82	Male	AB+	Asthma	01/07/2
ChRISTopher BerG	58	Female	AB-	Cancer	23/05/2
mIchElLe daniELs	72	Male	O+	Cancer	19/04/2
aaRon MARtiNeZ	38	Female	A-	Hypertension	13/08/2
connOR HANsEn	75	Female	A+	Diabetes	12/12/2
rObeRt bAuer	68	Female	AB+	Asthma	22/05/2
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MS. nAtalIE gAMble	46	Female	AB-	Obesity	01/01/2
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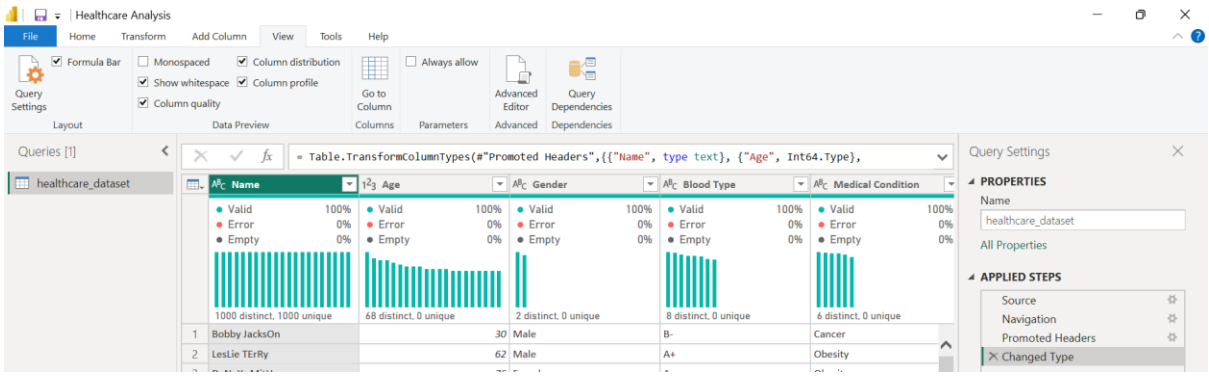
LuKE BuRgEss	34	Female	A-	Hypertension	04/03/2
dANIEL schmIdt	63	Male	B+	Asthma	15/11/2
tIMOTHY burNs	67	Female	A-	Asthma	28/06/2
ChRISToPHEr BRiGhT	48	Male	B+	Asthma	21/01/2

2. Data Cleaning

2.1 Missing Values

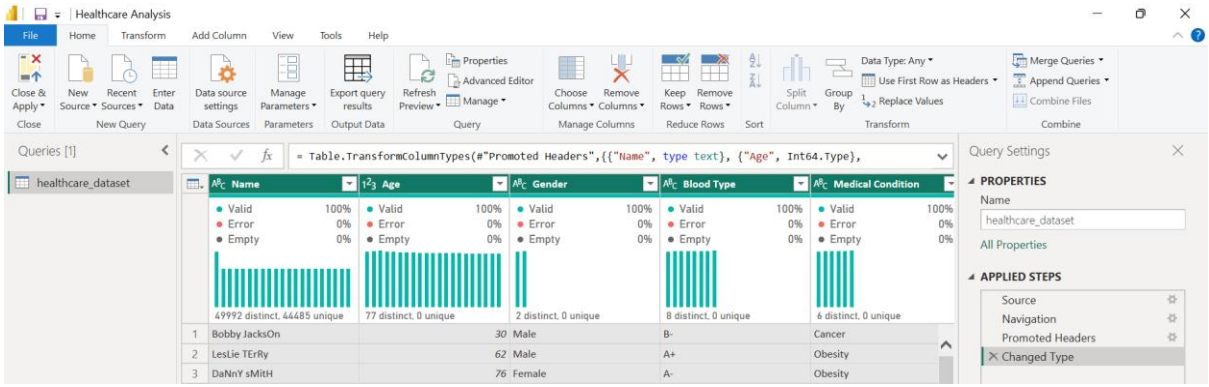
First, I checked for missing values, but since this dataset is synthetic, all 15 columns showed 100% valid and 0% empty, so there were no missing values.



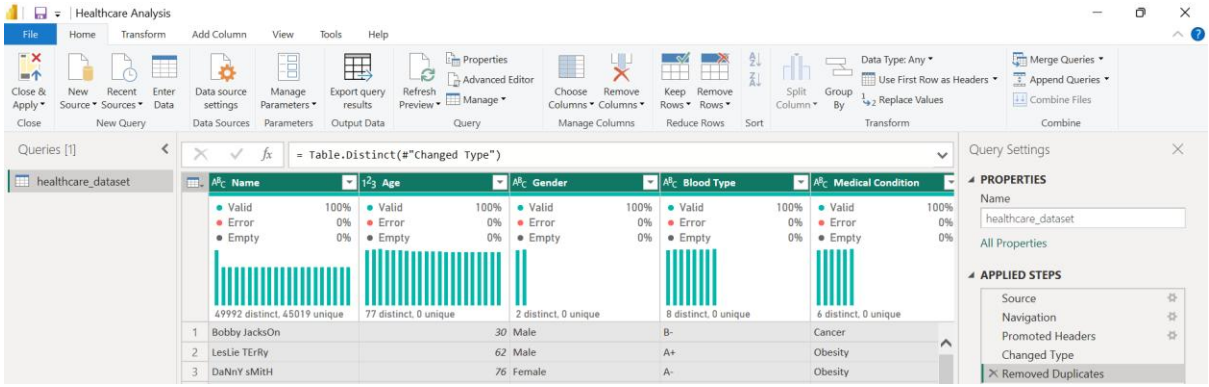
2.2 Removed Duplicates

I then proceeded to remove duplicates. I used all the columns to ensure that only identical records were removed, which meant that patients with multiple visits on different dates were kept, and only true duplicate rows were cleaned.

Before:

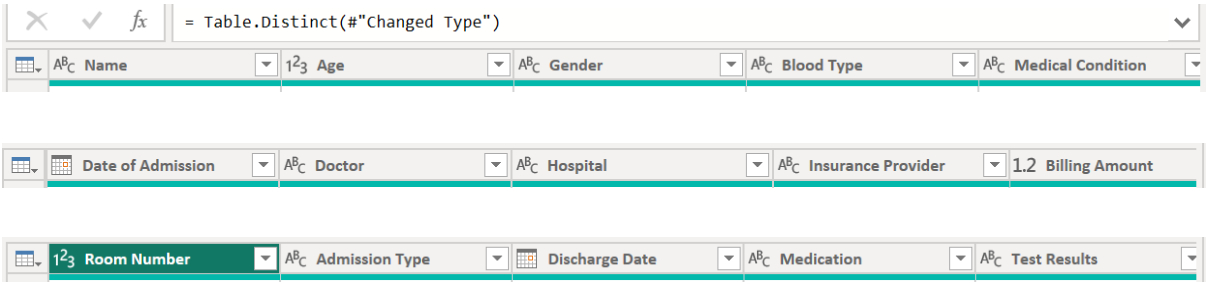


After:



2.3 Wrong Data Types

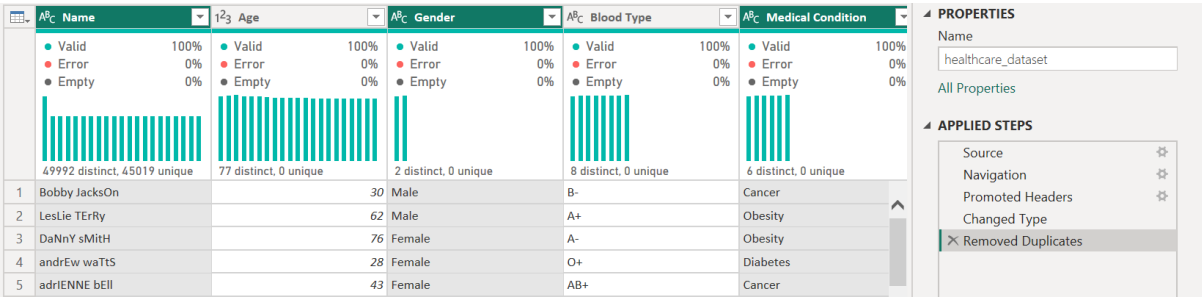
I checked the data types, and I confirmed they were already correctly assigned, so no major changes were required.



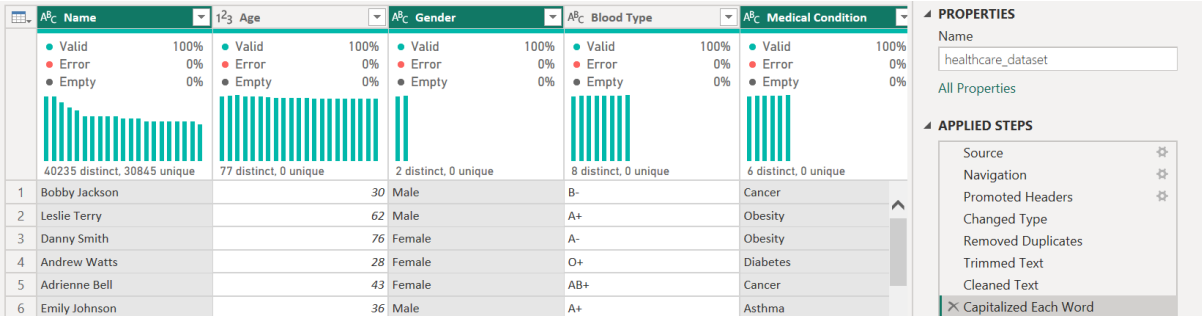
2.4 Inconsistent Category Names

I checked the columns: name, gender, medical condition, doctor, hospital, insurance provider, admission type, test results, and medications. I removed trailing spaces, non-printable characters, and ensured consistent casing.

Before:

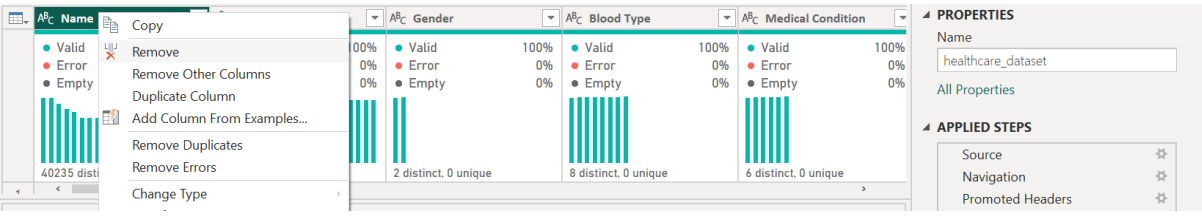


After:



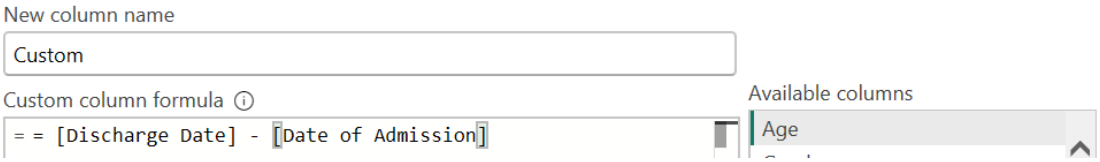
2.5 Unnecessary Columns

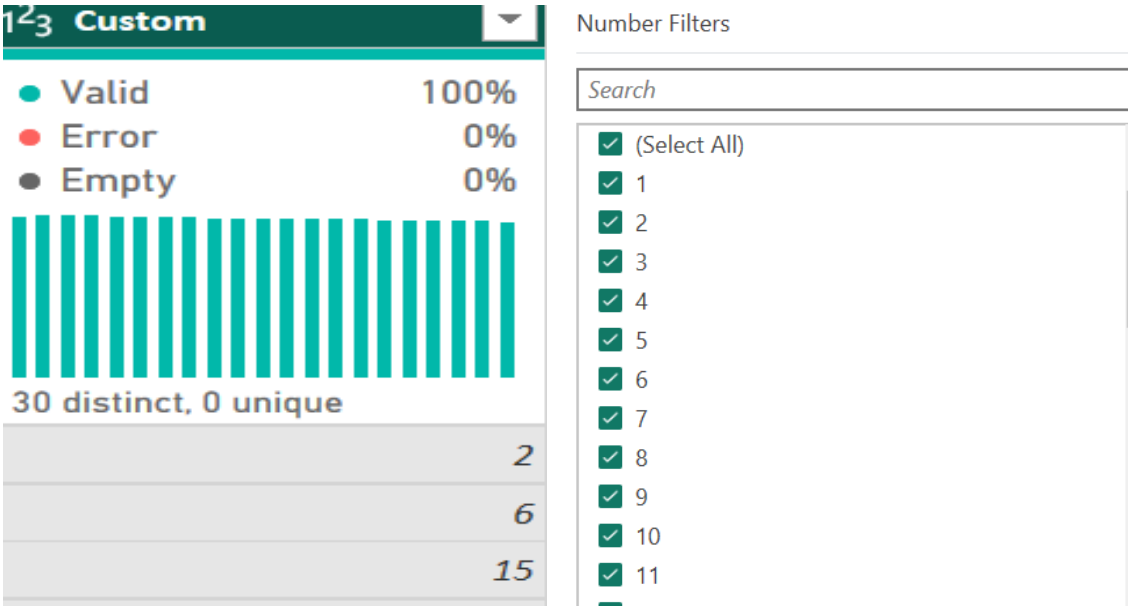
I removed the ‘Name’ column because the patient's name was just a personal identifier. It added no analytical value. It could not be aggregated or trended, also for privacy concerns, and would clutter visually if added.



2.6 Invalid records

I checked the columns for any invalid records, but since it was synthetic data, most of the records were valid. Though I went ahead and checked whether the admission dates and discharge dates were valid by subtracting the dates using a custom column, and all the records for the new custom column were positive, hence I verified the validity of the dates.



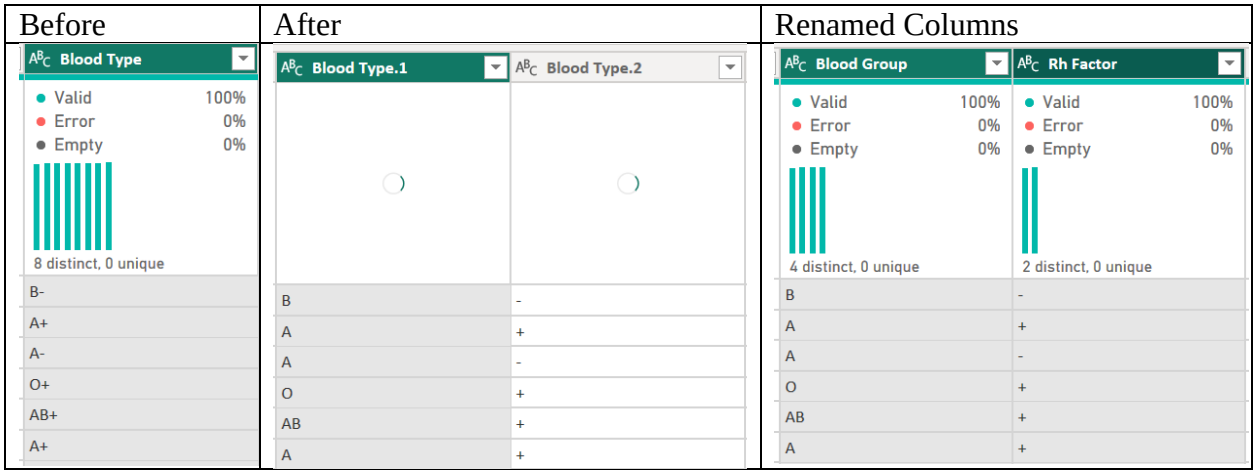


Also, since all the values were 1 or greater, it meant that the dataset specifically tracked inpatient stays rather than outpatient or ER visits. Also, I renamed the column 'LENGTH OF STAY' to use it later.

3. Data Transformation

3.1 Splitting Columns

I extracted the Blood Group and Rh Factor from the Blood Type column into two separate columns. This enables independent filtering and analysis by blood group category and Rh factor.

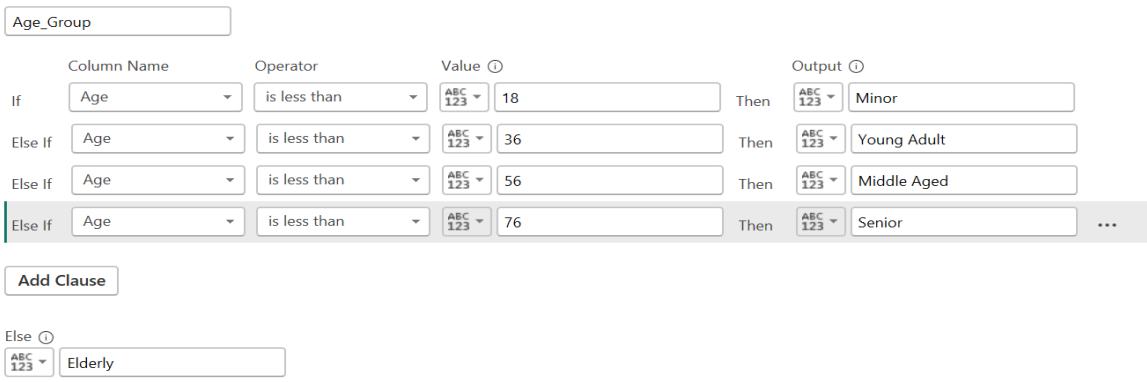


Before	After		Renamed Columns	
Blood Type	Blood Type.1	Blood Type.2	Blood Group	Rh Factor
Valid 100%			Valid 100%	Valid 100%
Error 0%			Error 0%	Error 0%
Empty 0%			Empty 0%	Empty 0%
8 distinct, 0 unique			4 distinct, 0 unique	2 distinct, 0 unique
B-	B	-	B	-
A+	A	+	A	+
A-	A	-	A	-
O+	O	+	O	+
AB+	AB	+	AB	+
A+	A	+	A	+

3.2 Conditional Columns

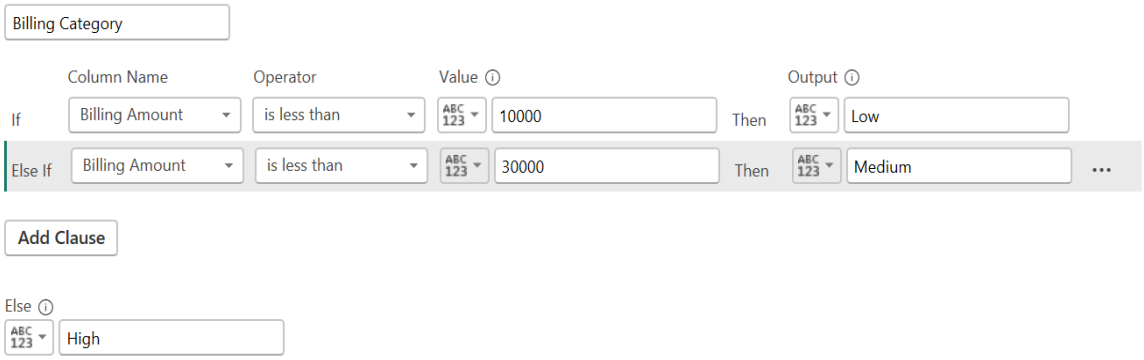
1. Age Group

I created a conditional column to group the raw age values into meaningful categories. The categories are: Minors, Young Adults, Middle Aged, Senior, and Elderly. This Grouping was to provide a clearer demographic segmentation.



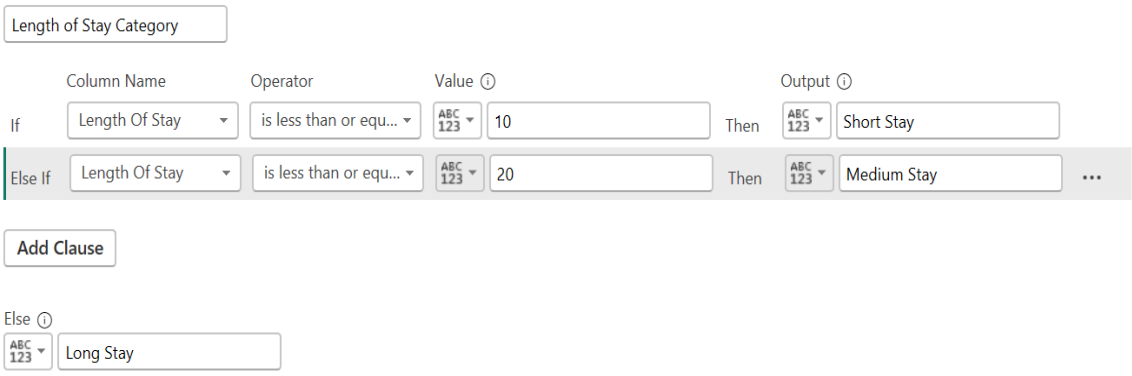
2. Billing Category

I first identified the negative values on the ‘Billing Amount’ column, filtered them out so that all the numbers to be greater than 0 as the negative values were inconsistent with real-world healthcare billing and were likely due to the synthetic data. Then I created the conditional column next and these were the ranges I chose:



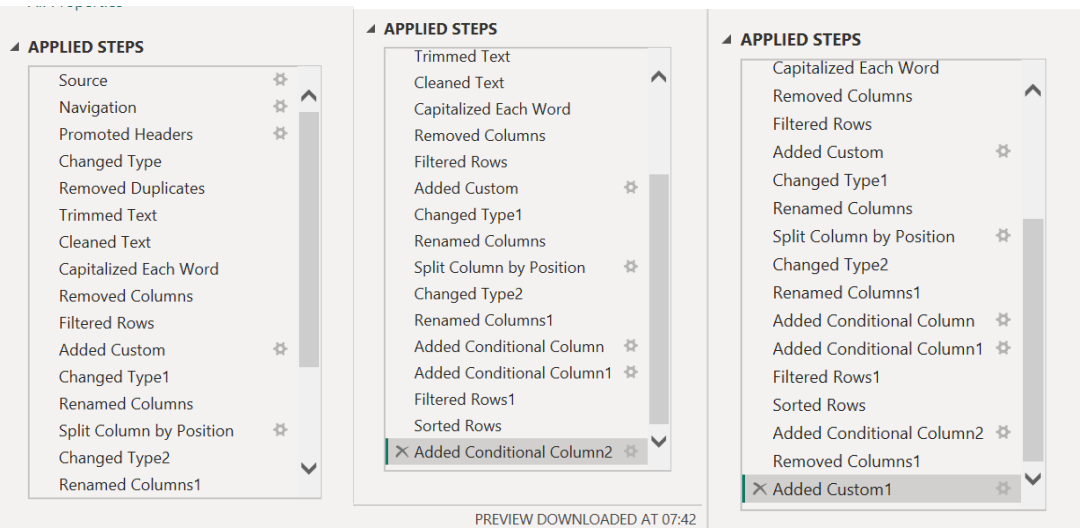
3. Length of stay

I created a conditional column to group the length of stay into short, medium, and long categories. This simplifies the data, making it easier to analyse patterns and compare patient stays in visuals.



3.3 Transformation Steps

After:



3.4 Why each transformation was necessary

- I removed duplicate rows to ensure data accuracy and avoid double counting in analysis.

- I checked for missing values to ensure data completeness and reliability of results.

- I verified data types to ensure correct calculations and prevent errors in analysis.

- I cleaned text fields by removing extra spaces and standardizing casing to ensure consistent categorization and accurate filtering.

- I removed the Name column because it was a unique identifier with no analytical value and would not contribute to aggregation or insights.

- I validated date fields and created a Length of Stay column to ensure data consistency and enable analysis of patient duration.

- I split the Blood Type column into Blood Group and Rh Factor to allow more detailed and flexible analysis.

- I created age groups to simplify demographic analysis and make patterns easier to interpret.

- I removed negative billing values because they are unrealistic in a healthcare context and would distort financial analysis.

- I grouped billing amounts into categories to simplify comparison and improve visualization.

- I categorized length of stay to make it easier to analyze patient duration patterns.

- I created a date dimension table to support time intelligence analysis and ensure a complete timeline.

Part C: Data Modelling

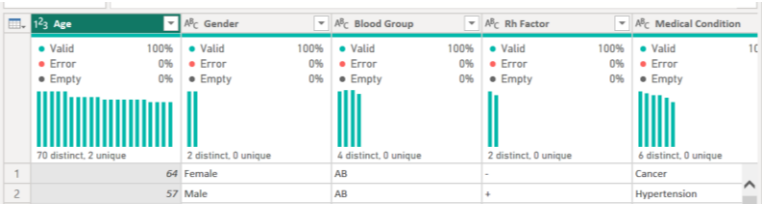
4.1 Why a Star Schema

I used a star schema because it separates transactional data (the fact table) from descriptive data (dimension tables). I structured my dataset into one fact table and multiple dimension tables to implement this model.

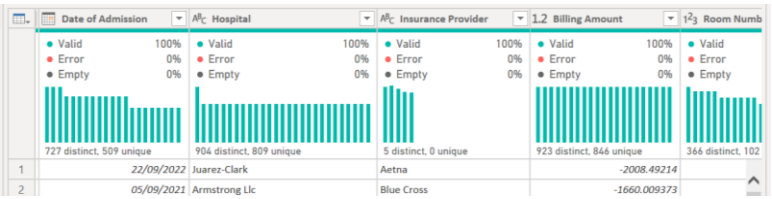
4.2 Fact_Admissions

I used this as the central fact table because it contains transactional data such as admissions, billing amount, and length of stay.

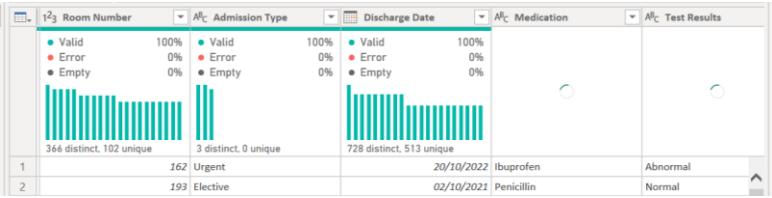
Columns I kept:



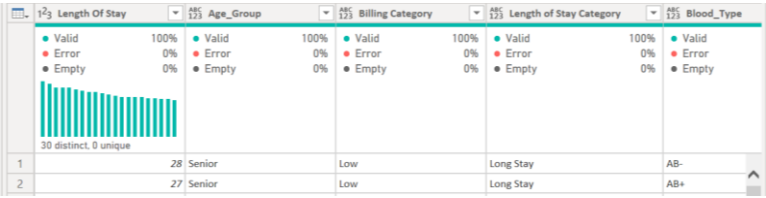
	Age	Gender	Blood Group	Rh Factor	Medical Condition
	Valid 100%, Error 0%, Empty 0%	Valid 100%, Error 0%, Empty 0%	Valid 100%, Error 0%, Empty 0%	Valid 100%, Error 0%, Empty 0%	Valid 10, Error, Empty
	70 distinct, 2 unique	2 distinct, 0 unique	4 distinct, 0 unique	2 distinct, 0 unique	6 distinct, 0 unique
1	64	Female	AB	-	Cancer
2	57	Male	AB	+	Hypertension



	Date of Admission	Hospital	Insurance Provider	Billing Amount	Room Numb
	Valid 100%, Error 0%, Empty 0%	Valid 100%, Error 0%, Empty 0%	Valid 100%, Error 0%, Empty 0%	Valid 100%, Error 0%, Empty 0%	Valid, Error, Empty
	727 distinct, 509 unique	904 distinct, 809 unique	5 distinct, 0 unique	923 distinct, 846 unique	366 distinct, 102
1	22/09/2022	Juarez-Clark	Aetna	-2008.49214	
2	05/09/2021	Armstrong Llc	Blue Cross	-1660.009373	



	Room Number	Admission Type	Discharge Date	Medication	Test Results
	Valid 100%, Error 0%, Empty 0%	Valid 100%, Error 0%, Empty 0%	Valid 100%, Error 0%, Empty 0%		
	366 distinct, 102 unique	3 distinct, 0 unique	728 distinct, 513 unique		
1	162	Urgent	20/10/2022	Ibuprofen	Abnormal
2	193	Elective	02/10/2021	Penicillin	Normal

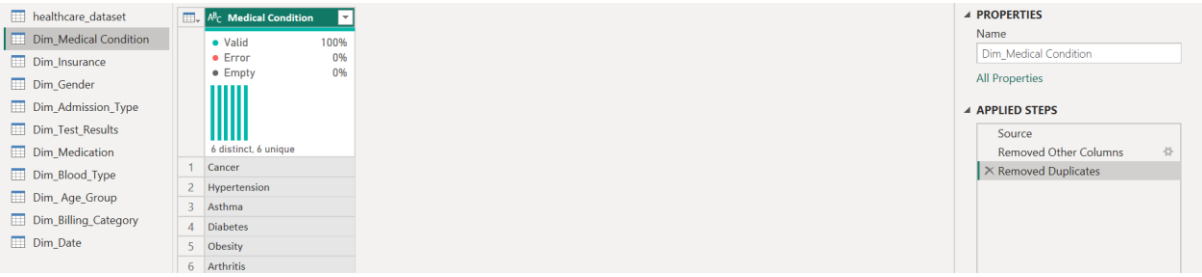


	Length Of Stay	Age_Group	Billing Category	Length of Stay Category	Blood_Type
	Valid 100%, Error 0%, Empty 0%	Valid 100%, Error 0%, Empty 0%	Valid 100%, Error 0%, Empty 0%	Valid 100%, Error 0%, Empty 0%	Valid, Error, Empty
	30 distinct, 0 unique				
1	28	Senior	Low	Long Stay	AB-
2	27	Senior	Low	Long Stay	AB+

4.3 Building the Dimension Tables in Power Query

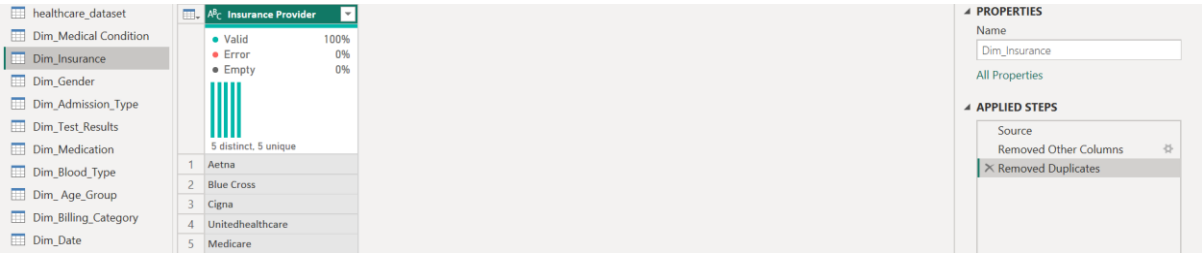
1. Dim_Medical_Conditions

I used this to store different medical conditions for filtering and comparison.



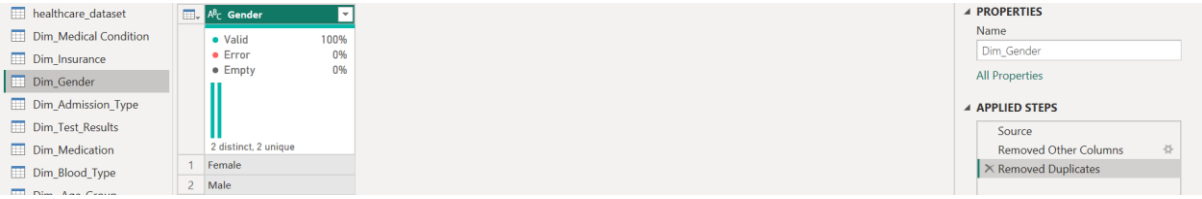
2. Dim_Insuarance

I created this to categorize patients based on insurance providers for financial analysis.



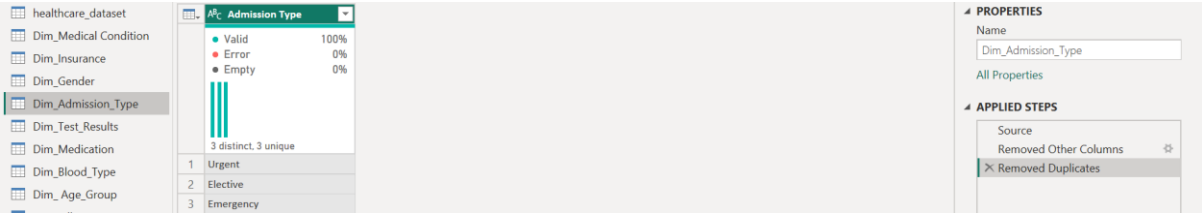
3. Dim_Gender

I used this to enable demographic analysis based on gender.



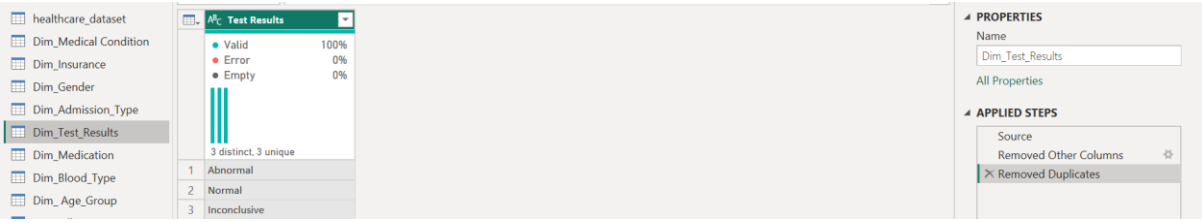
4. Dim_Admission_Types

I created this to analyze different types of admissions such as emergency and elective.



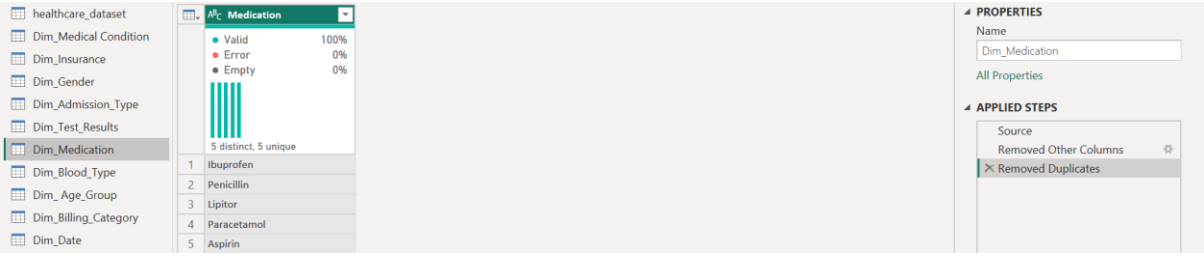
5. Dim_Test_Results

I used this to track and analyze patient test outcomes.



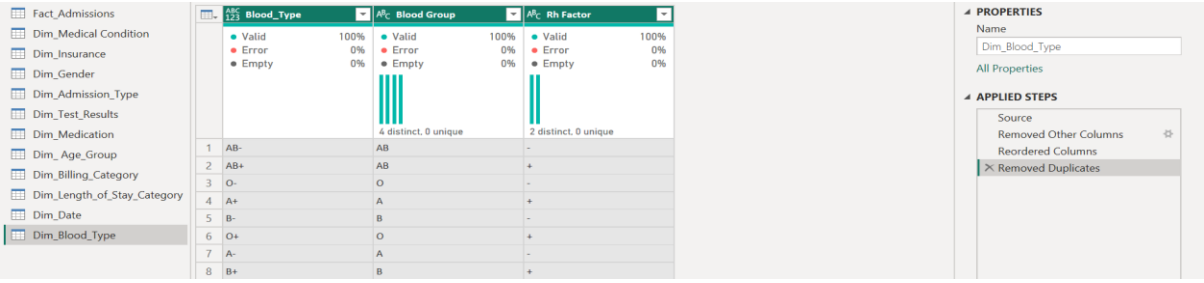
6. Dim_Medications

I created this to analyze medication usage patterns.



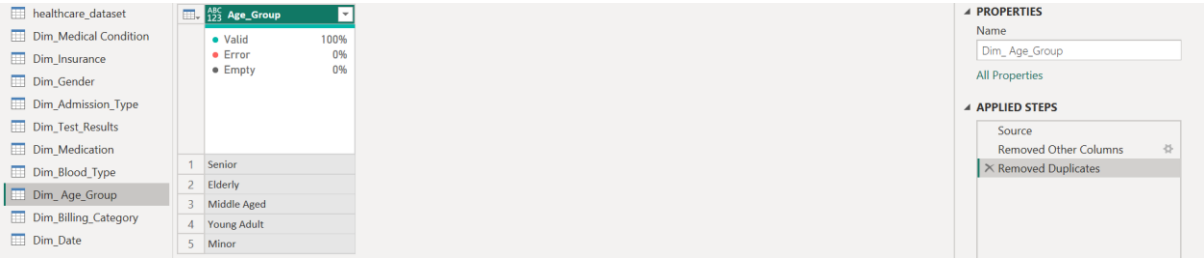
7. Dim_Blood_Type

I used this to categorize patients by blood type for clinical analysis.



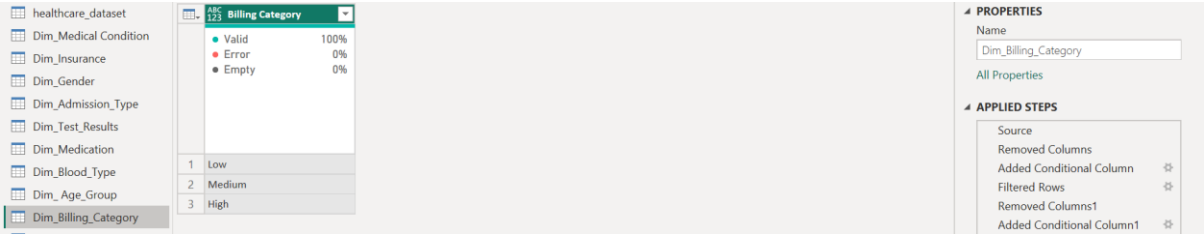
8. Dim_Age_Group

I created this to group patients into age categories for easier demographic insights.



9. Dim_Billing_Category

I used this to group billing amounts into categories for simplified financial analysis.

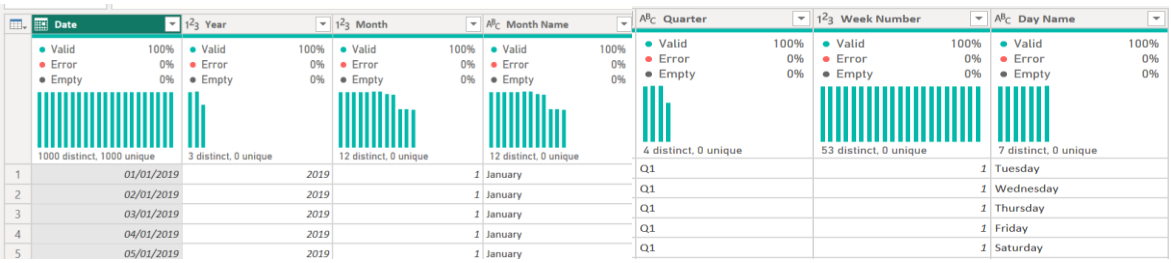


10. Dim_Date

I created a Date dimension table using Power Query code to generate a complete calendar and support time intelligence, instead of extracting date parts manually. I created this table to store date attributes such as year, month, and quarter to support time-based analysis.

```
let
    StartDate = #date(2019, 1, 1),
    EndDate = #date(2024, 12, 31),
    DateList = List.Dates(StartDate, Duration.Days(EndDate - StartDate) + 1, #duration(1,0,0,0)),
    TableFromList = Table.FromList(DateList, Splitter.SplitByNothing(), {"Date"}),
    ChangedType = Table.TransformColumnTypes(TableFromList, {{"Date", type date}}),
    AddYear = Table.AddColumn(ChangedType, "Year", each Date.Year([Date]), Int64.Type),
    AddMonth = Table.AddColumn(AddYear, "Month", each Date.Month([Date]), Int64.Type),
    AddMonthName = Table.AddColumn(AddMonth, "Month Name", each Date.ToText([Date], "MMMM"), type text),
    AddQuarter = Table.AddColumn(AddMonthName, "Quarter", each "Q" & Text.From(Date.QuarterOfYear([Date])), type text),
    AddWeek = Table.AddColumn(AddQuarter, "Week Number", each Date.WeekOfYear([Date]), Int64.Type),
    AddDayName = Table.AddColumn(AddWeek, "Day Name", each Date.ToText([Date], "dddd"), type text)
in
    AddDayName
```

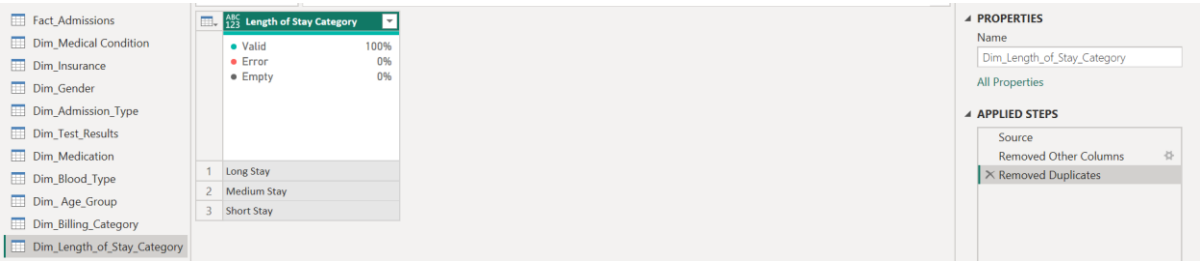
Tables:



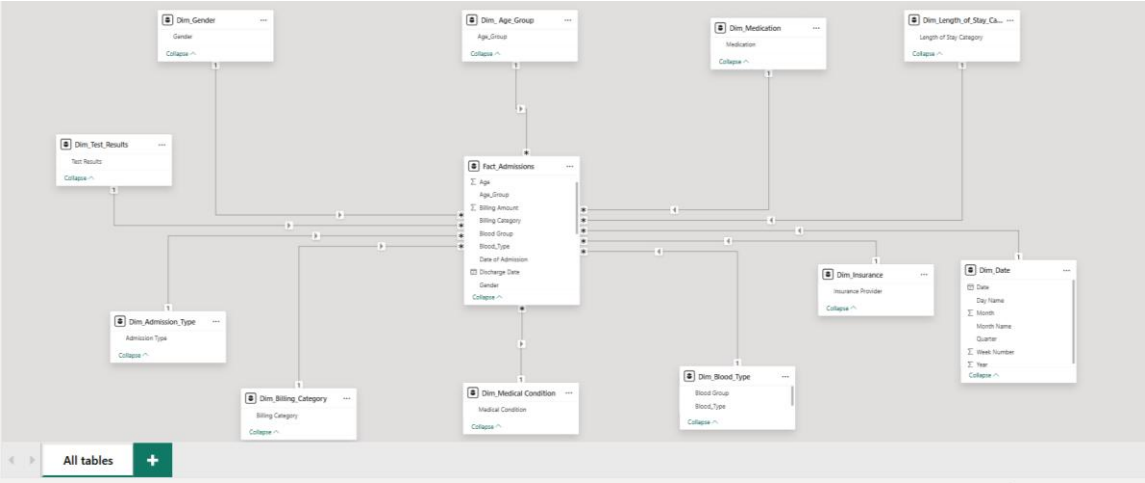
	Date	Year	Month	Month Name	Quarter	Week Number	Day Name
1	01/01/2019	2019	1	January	Q1	1	Tuesday
2	02/01/2019	2019	1	January	Q1	1	Wednesday
3	03/01/2019	2019	1	January	Q1	1	Thursday
4	04/01/2019	2019	1	January	Q1	1	Friday
5	05/01/2019	2019	1	January	Q1	1	Saturday

11. Dim_Length_of_Stay

I created this to categorize patient stay duration into short, medium, and long stays.

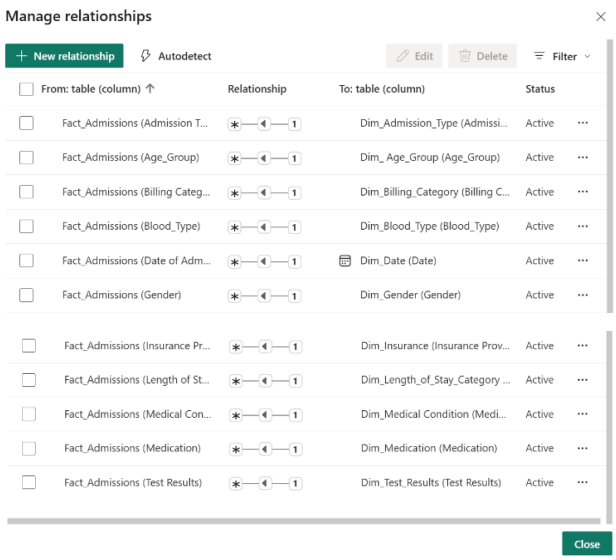


4.4 Final Star Schema Diagram



The model structure supports efficient reporting because I used a star schema, where the fact table is connected to multiple dimension tables. This reduces data complexity, improves performance by minimizing data scanning, and allows filters from dimension tables to efficiently slice the fact table. It also enables faster and more accurate DAX calculations, making the reports more responsive and easier to analyze.

4.5 Relationships in Model View



I created relationships between the fact table and dimension tables using a many-to-one (*:1) cardinality, where each dimension table contains unique values. The relationships are single-direction, allowing filters from dimension tables to flow to the fact table. This ensures efficient filtering, avoids ambiguity, and supports accurate analysis.

4.6 Why I Think the Model Structure is Efficient for Reporting

I think this model structure is efficient because it keeps things simple, fast, and reliable when building reports in Power BI.

1. It makes queries faster

Every interaction in the report only goes one step from a dimension table to the fact table. There are no complicated joins or multiple paths for Power BI to figure out. Hence when I filter something like a medical condition or age group, the system responds quickly even when multiple visuals are on the page.

2. It makes calculations more accurate

The structure ensures filters behave in a clear and predictable way which is important for measures like YTD billing or year-on-year growth, because they depend on correct date filtering. With this setup, I don't have to worry about Power BI mixing up filters or returning incorrect results silently.

3. Filters work independently

Each slicer comes from its own separate dimension table, so they don't interfere with each other. For example, filtering by age group, condition, and insurance all work together properly, instead of one filter breaking or overriding another. This makes the dashboard more reliable and easier to use.

4. It is easy to update and scale

I like this structure because it's flexible. If new data comes in, like a new insurance provider or a new medical condition it will mean that I only update the relevant dimension table. I don't need to rebuild the whole model or change the fact table, which saves a lot of time and effort.

5. It makes reporting easier to understand

Instead of raw data values, users see clean labels like age groups, billing categories, and months. This makes the report more user-friendly and reduces confusion when building or interpreting visuals.

Part D: DAX Measures and Calculated Columns

I created all DAX measures in a dedicated **_Measures table**, which I added to the model. I stored all measures there instead of spreading them across different tables. I did this to make it easier to locate, manage, and reuse measures during report development.

DAX Measures

1. Total Billing Amount

I created this measure using the SUM function on the Billing Amount column in the Fact_Admissions table. It calculates the total revenue generated from all admissions and automatically adjusts based on filters applied in the report. I used this as a key KPI because it represents the overall financial performance of the hospital system and is one of the most important metrics for decision-making.

```
Total Billing Amount =
SUM(Fact_Admissions[Billing Amount])
```

2. Average Billing per Admission

I created this measure using AVERAGE on the Billing Amount column. It calculates the average cost per admission, regardless of total volume. I included this because it allows fair comparison across different conditions or hospitals, especially when patient volumes are not equal.

```
Average Billing per Admission =
AVERAGE(Fact_Admissions[Billing Amount])
```

3. Total Admissions

I created this measure using COUNTROWS on the Fact_Admissions table. It counts each admission record and gives the total number of patient admissions in the current filter context. I used this to understand patient volume and workload across hospitals and conditions.

```
Total Admissions =
COUNTROWS(Fact_Admissions)
```

4. % Abnormal Test Results

I created this measure using CALCULATE with a filter for “Abnormal” test results and divided it by total admissions using DIVIDE. This shows the proportion of abnormal results out of all admissions. I included it as a clinical indicator to help identify patterns in patient health severity.

```
% Abnormal Test Results =
DIVIDE(
    CALCULATE(
        COUNTROWS(Fact_Admissions),
        Dim_Test_Results[Test Results] = "Abnormal"
    ),
    [Total Admissions],
    0
)
```

5. Year-to-Date (YTD) Billing

I created this measure using TOTALYTD on the Billing Amount column with the Dim_Date table. It calculates cumulative billing from the start of the year to the current date in context. I used this to track financial performance over time and monitor progress within a year.

```
YTD Billing =
TOTALYTD(
    SUM(Fact_Admissions[Billing Amount]),
    Dim_Date[Date]
)
```

6. Previous Year Billing

I created this measure using CALCULATE with SAMEPERIODLASTYEAR. It shifts the date context back by one year and recalculates total billing. I used this to enable year-over-year comparison in my report visuals.

```
Previous Year Billing =
CALCULATE(
    [Total Billing Amount],
    SAMEPERIODLASTYEAR(Dim_Date[Date])
)
```

7. Billing Growth %

I created this measure by comparing current billing to previous year billing and dividing the difference using DIVIDE. This gives the percentage growth in revenue. I used it to evaluate financial performance trends over time.

```
Billing Growth % =
DIVIDE(
    [Total Billing Amount] - [Previous Year Billing],
    [Previous Year Billing],
    0
)
```

8. Average Length of Stay

I created this measure using AVERAGE on the Length of Stay column. It calculates the average number of days patients stay in hospital. I included this because it is a key operational efficiency metric.

```
Average Length of Stay =
AVERAGE(Fact_Admissions[Length of Stay])
```

9. Top Hospital Rank

I created this measure using RANKX to rank hospitals based on total billing. I used ALL to ensure ranking is done across the full dataset and not affected by filters. I included this to identify top and bottom performing hospitals.

```
Top Hospital Rank =
RANKX(
    ALL(Fact_Admissions[Hospital]),
    [Total Billing Amount],
    ,
    DESC,
    DENSE
)
```

10. Billing % of Total

I created this measure using DIVIDE, where each category's billing is divided by the total billing across all admissions. I used ALL inside CALCULATE to remove filters and return the true total. This helps show which categories contribute the most to overall revenue.

```
Billing % of Total =
DIVIDE(
    [Total Billing Amount],
    CALCULATE(
        [Total Billing Amount],
        ALL(Fact_Admissions)
    ),
    0
)
```

11. Previous Year Admissions

I created this measure using the CALCULATE function combined with SAMEPERIODLASTYEAR, referencing the Date column in the Dim_Date table. It takes the Total Admissions measure and recalculates it for the equivalent period twelve months prior. I used this as a supporting measure because on its own the Total Admissions number has no context; a stakeholder cannot tell whether 54,966 admissions is an improvement or a decline without knowing what last year's figure was. This measure provides that baseline and feeds directly into the Admissions Growth % calculation.

```
Previous Year Admissions =
CALCULATE(
    [Total Admissions],
    SAMEPERIODLASTYEAR(Dim_Date[Date])
)
```

12. Admissions Growth %

I created this measure using the DIVIDE function, subtracting Previous Year Admissions from Total Admissions and dividing the result by Previous Year Admissions. DIVIDE was used instead of a standard division operator because it handles blank or zero denominators safely, returning blank instead of an error when prior year data does not exist. I used this measure to directly answer the business question of whether patient admission volumes are growing or shrinking year on year. It is displayed as a delta chip beneath the Total

Admissions KPI card and is formatted as a percentage with conditional formatting applied so it shows green when positive and red when negative.

```
Admissions Growth % =
DIVIDE(
    [Total Admissions] - [Previous Year Admissions],
    [Previous Year Admissions]
)
```

13. Top Medication

I created this measure using the TOPN function nested inside MAXX, referencing the Medication column in the Dim_Medication table. TOPN ranks all medications by their Total Admissions count and returns only the first row which means the medication associated with the highest number of admissions. MAXX then extracts the medication name from that result so it displays as text on the dashboard. I used this measure to answer the business question of which medication is most frequently prescribed across the hospital system.

```
Top Medication =
MAXX(
    TOPN(
        1,
        VALUES(Dim_Medication[Medication]),
        CALCULATE([Total Admissions]),
        DESC
    ),
    Dim_Medication[Medication]
)
```

14. Prior Year YTD Billing

I created this measure using CALCULATE combined with SAMEPERIODLASTYEAR, wrapping it around my existing YTD Billing measure. Rather than returning the full prior year total, it specifically returns the cumulative billing up to the same point in the previous year, so if the current date is the end of Q2, it returns what billing looked like at the end of Q2 last year.

```
Prior Year YTD Billing =
CALCULATE(
    [YTD Billing],
    SAMEPERIODLASTYEAR(Dim_Date[Date])
)
```

15. YTD vs Prior Year %

I created this measure using the DIVIDE function, subtracting Prior Year YTD Billing from YTD Billing and dividing the result by Prior Year YTD Billing. I used DIVIDE instead of standard division because it handles situations where the prior year figure is blank or zero without returning an error, which can happen early in the year before sufficient historical data exists.

```
YTD vs Prior Year % =
DIVIDE(
    [YTD Billing] - [Prior Year YTD Billing],
    [Prior Year YTD Billing]
)
```

16. On Target Pace

I created this measure using an IF statement that evaluates the YTD vs Prior Year % measure and returns either "On Track" or "Behind" depending on whether the result is zero or above versus negative. It is the final layer in a three-measure chain, YTD Billing feeds Prior Year YTD Billing, which feeds YTD vs Prior Year %, which feeds this status flag.

```
On Target Pace =
IF(
    [YTD vs Prior Year %] >= 0,
    "On Track",
    "Behind"
)
```

17. Top Insurance

I created this measure using TOPN nested inside MAXX, referencing the Insurance Provider column in Dim_Insurance. TOPN ranks all insurance providers by Total Billing Amount in descending order and returns only the top ranked result. MAXX then extracts the provider name from that result so it displays as readable text on a Card visual rather than a number.

```
Top Insurance Provider =
MAXX(
    TOPN(
        1,
        VALUES(Dim_Insurance[Insurance Provider]),
        CALCULATE([Total Billing Amount]),
        DESC
    ),
    Dim_Insurance[Insurance Provider]
)
```

Calculated Columns

Calculated columns were created when I needed values at the individual row level rather than aggregated results. Unlike measures, calculated columns are computed once during data loading and stored in the table.

Risk Category

I created this calculated column using the SWITCH function to classify each admission based on test results and admission type. It assigns categories such as High Risk, Medium Risk, Monitor, and Low Risk depending on clinical conditions. I created this because the risk level needs to exist at row level so it can be used for slicing, grouping, and detailed analysis in visuals.

```
Risk Category =
SWITCH(
    TRUE(),
    Fact_Admissions[Test Results] = "Abnormal"
        && Fact_Admissions[Admission Type] = "Emergency",
    "High Risk",
    Fact_Admissions[Test Results] = "Abnormal"
        && Fact_Admissions[Admission Type] = "Urgent",
    "Medium Risk",
    Fact_Admissions[Test Results] = "Abnormal",
    "Low-Medium Risk",
    Fact_Admissions[Test Results] = "Inconclusive",
    "Monitor",
    "Low Risk"
)
```

How I Think the Measures Support Decision-Making

1. Total Billing Amount

I think this is the most important financial KPI in the entire dashboard because it gives a clear overall picture of revenue generated across all admissions. I used it as a baseline for all financial comparisons. Without this measure, I felt there would be no reference point for understanding financial performance in the dataset.

2. Average Billing per Admission

I included this measure because I think total billing alone can be misleading, since higher admission volumes naturally produce higher totals. This measure allows me to compare costs fairly across hospitals and conditions. I think it is especially useful for identifying whether certain conditions are more expensive per patient, regardless of how many patients are treated.

3. Average Length of Stay

I chose this measure because I think length of stay is a key indicator of hospital efficiency. From my perspective, longer stays increase costs and reduce bed availability, so this metric helps highlight inefficiencies in patient flow or treatment processes. I think it is particularly useful for comparing operational performance across hospitals and conditions.

4. % Abnormal Test Results

I included this measure because I think it provides important insight into patient health severity and diagnostic outcomes. I felt it was useful for identifying patterns where a higher proportion of abnormal results might indicate late patient presentation, worsening health conditions, or differences in diagnostic quality across hospitals.

5.Billing Growth %

I think this measure is essential for understanding financial performance over time rather than in isolation. It allows me to see not just whether revenue increased or decreased, but by how much compared to the previous year. I felt this was important for evaluating real growth trends and supporting financial forecasting decisions.

6. Previous Year Billing

I created this measure because I think year-over-year comparison is not possible without a clear historical baseline. It allows me to directly compare current performance with the same period last year. I used it as the foundation for the Billing Growth % measure.

7. YTD Billing

I included this measure because I think cumulative performance is more meaningful than isolated monthly values. It shows the financial trajectory of the hospital throughout the year, allowing me to assess whether performance is on track toward annual targets at any given point in time.

8. Total Admissions

I think this is a core operational metric because it shows patient volume across the entire dataset. It helps me understand workload distribution across hospitals, conditions, and time periods. I felt it was important for identifying demand patterns and areas where resources may be stretched.

9. Top Hospital Rank

I created this measure because I think ranking hospitals makes performance comparisons much clearer. Instead of manually sorting values, this measure automatically highlights the highest and lowest performing hospitals. I used it to support visual comparisons and focus attention on key performers and underperformers.

10. Billing % of Total

I included this measure because I think totals alone do not show proportional impact. It allows me to see which categories contribute most to overall revenue. I felt this was important for identifying cost drivers that have a disproportionate effect on total billing, even if they do not have the highest number of admissions.

11. Previous Year Admissions

This measure is not displayed directly on the dashboard but it is the foundational baseline that makes Admissions Growth % possible. Every year-on-year admission comparison in the report depends on this figure being calculated accurately. By isolating it as its own measure I also make it available for other visuals, such as a side-by-side line chart comparing current year and prior year admissions monthly, without having to rewrite the logic each time.

12. Admission Growth

The Total Admissions KPI card on its own only tells administrators how many patients were admitted, it gives no indication of whether that represents an improvement or a decline. Displaying this measure as a delta beneath the Total Admissions card means administrators can see both the volume and the direction of change simultaneously. This directly answers the business question of whether patient admissions have grown or shrunk year on year and provides the evidence base for capacity planning and staffing decisions.

13. Top Medication

While a bar chart can show the full medication distribution, surfacing the top medication as a single card value gives clinical administrators an immediate answer without needing to read and compare a full chart. More importantly it responds dynamically to slicers, filtering by condition, or age group, updates the card to show the top medication for that specific patient segment, which supports targeted clinical decision making rather than just a static headline figure.

14. Previous Year YTD Billing

Without this measure, comparing current YTD billing against a full prior year total would be misleading, administrators would always appear behind target in the first half of the year simply because the year is not finished. This measure creates a fair like-for-like comparison at every point in time, giving administrators an accurate and honest baseline to measure current financial pace against.

15. YTD vs Prior Year %

A raw billing figure tells administrators nothing about performance trajectory on its own. This measure converts two numbers into a single percentage that immediately answers whether current cumulative revenue is ahead of or behind where it was at this point last year. It also feeds directly into the On-Target Pace status flag, making it a foundational calculation for the entire financial monitoring section of Page 3.

16. On Target Pace

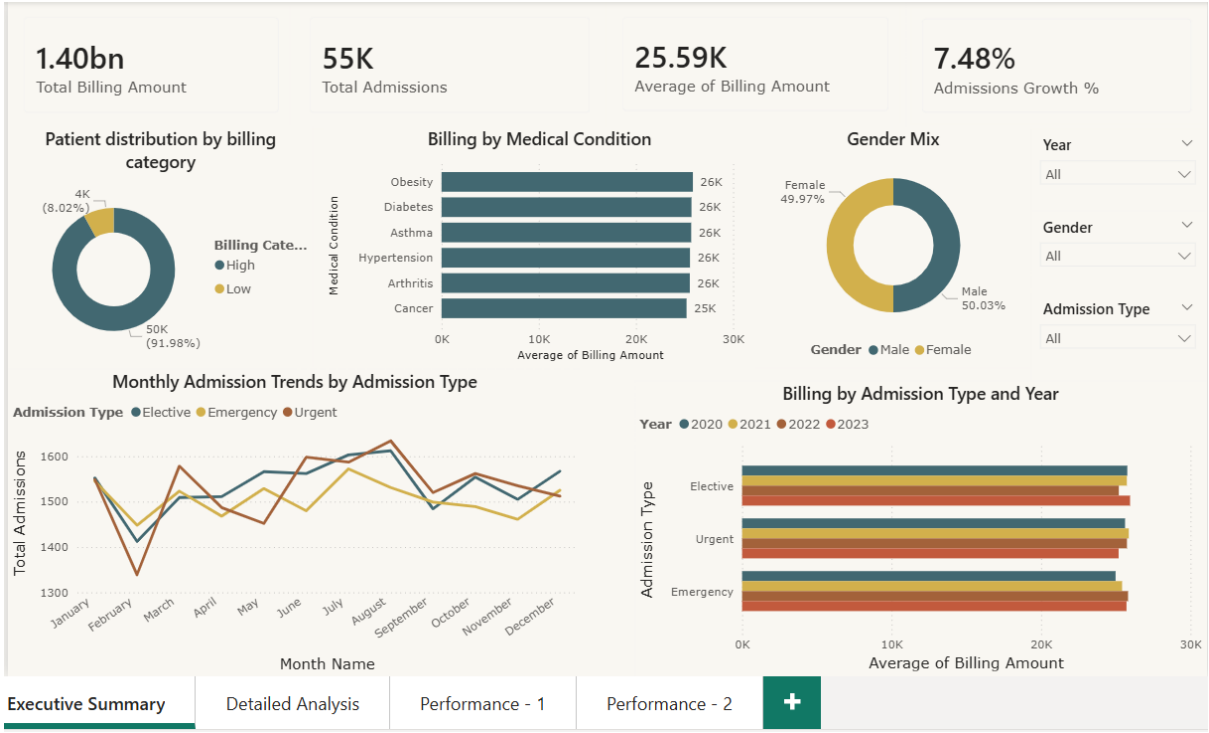
Not every stakeholder who reads the dashboard will want to interpret a percentage. A finance director might, but a ward administrator or clinical lead may simply need to know whether the hospital is performing above or below last year's pace at a glance. By converting the percentage into a plain text outcome this measure makes the dashboard accessible to a wider audience. Displayed as a Card visual.

17. Top Insurance

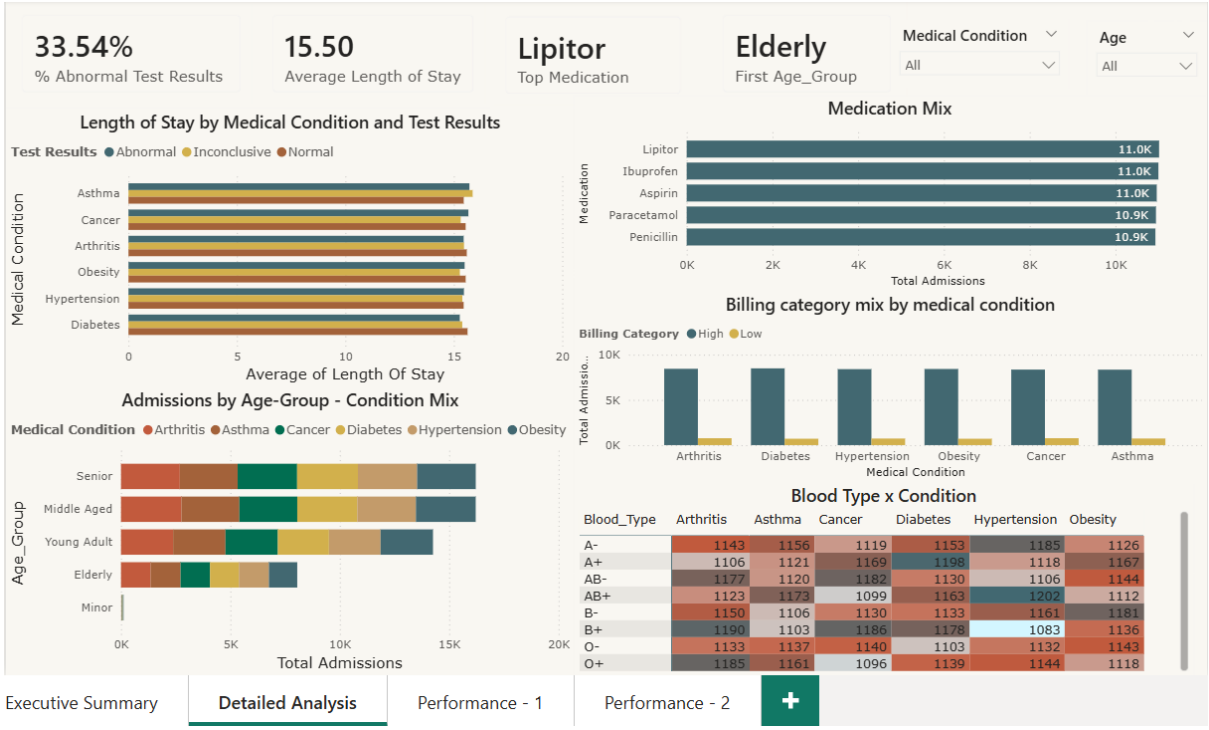
Knowing which insurance provider is generating the most billing at any point in time is directly relevant to contract management and revenue planning. Because the measure responds dynamically to slicers, filtering the dashboard by medical condition or admission type will update the card to show the top insurer for that specific patient segment

Part E: Dashboard Design and Visualization

Page 1: Executive Summary

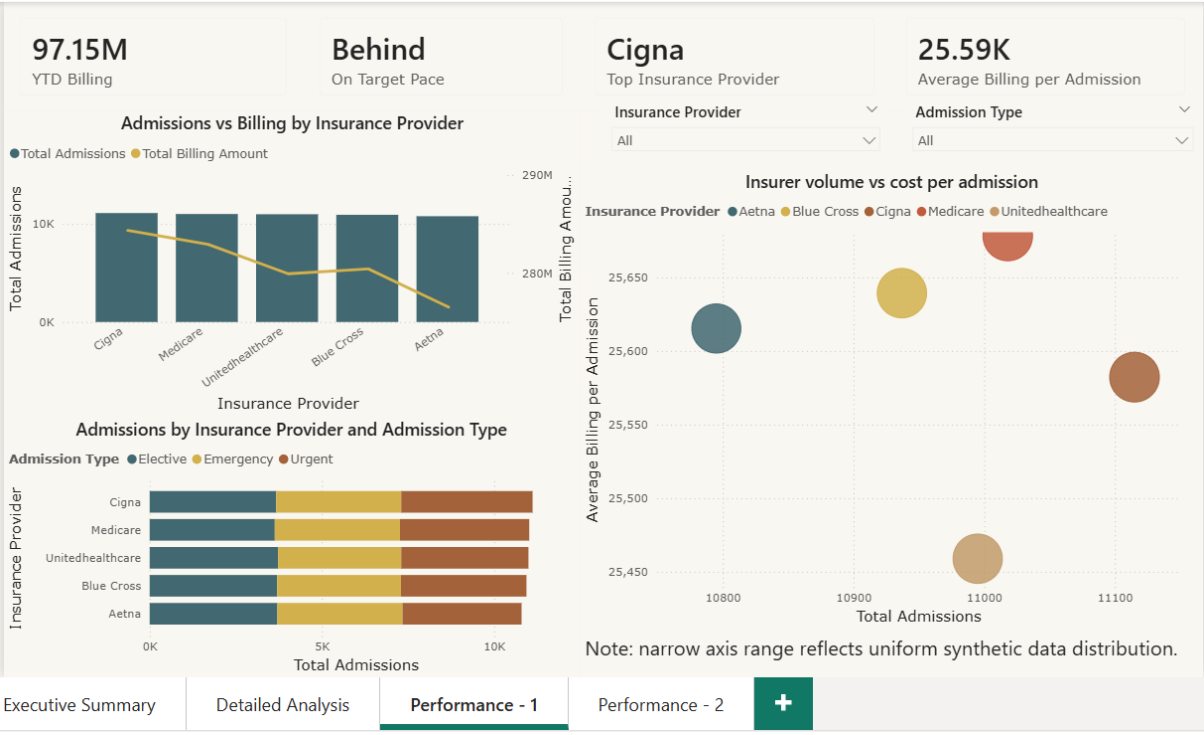


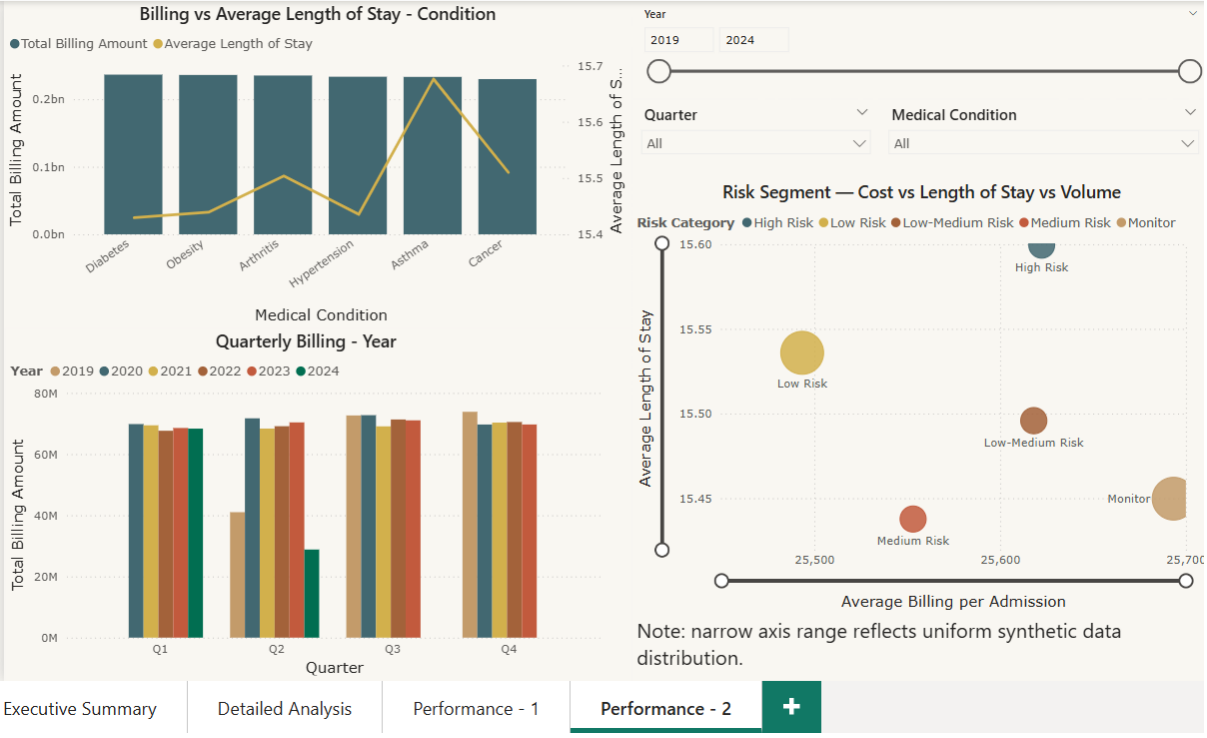
Page 2: Detailed Analysis breakdown view



Blood_Type	Arthritis	Asthma	Cancer	Diabetes	Hypertension	Obesity
A-	1143	1156	1119	1153	1185	1126
A+	1106	1121	1169	1198	1118	1167
AB-	1177	1120	1182	1130	1106	1144
AB+	1123	1173	1099	1163	1202	1112
B-	1150	1106	1130	1133	1161	1181
B+	1190	1103	1186	1178	1083	1136
O-	1133	1137	1140	1103	1132	1143
O+	1185	1161	1096	1139	1144	1118

Page 3 & 4: Performance Monitoring





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Design Choices

When I was building this dashboard, I made a deliberate decision to organise it across four pages using what I would call a progressive disclosure approach. The idea was that each page should answer a different level of question for a different type of user.

Page 1 is built for senior leadership who need to understand the hospital's current state in under ten seconds hence why I put four headlines ‘KPIs’at the top and arranged the six charts below to answer the basic what, when, and who questions. As you move through Pages 2, 3, and 4 the analysis gets progressively deeper, targeting operational managers, payer-relationship owners, and financial analysts respectively. A senior leader can stop at Page 1 and still leave with a complete picture. An analyst continues through for the root-cause detail.

One of the most intentional structural decisions I made was the KPI strip at the top of every page. I kept the position, typography, and colour treatment identical across all four pages on purpose. The reason is that when users see the same element in the same place every time they stop having to hunt for context and can focus on what the numbers are actually telling them. I also made a deliberate choice about how to display performance on those cards, on Page 1 the Admissions Growth % card shows a percentage because growth is a numerical concept, but on Page 3 the On Target Pace card shows the word "Behind" rather than a bare number because I felt that target attainment is more meaningfully communicated as a category than as a figure someone has to interpret.

Every chart type I chose was driven by the shape of the question I was trying to answer rather than by visual preference. I used horizontal bar charts for all the ranking questions, conditions by billing, insurers by volume, because research shows that the eye reads length more accurately than angle or area. I used line charts for all the time-series questions because lines communicate direction and rate of change better than any other chart type. I used the donut chart exactly once on Page 1 for the gender split because it is a genuine two-part ratio and donut charts become unreadable above three or four categories so I did not use one anywhere else. I reserved scatter plots for the correlation questions which was insurer volume versus cost per admission, and risk segment cost versus length of stay, because those are the only questions where you genuinely need to compare two continuous variables at the same time.

I kept the colour palette deliberately restrained throughout. I used one teal as the primary data colour across all bars and only introduced supporting colours: mustard, orange, brown. This

was a conscious design choice because a visually calm dashboard surfaces the data rather than competing with it, and it means users learn what each colour signals and carry that understanding across the whole report.

The layout follows a consistent grid across all four pages, KPI strip on top, slicers on the right, primary content in a two or three column arrangement below. That consistency is what allows users to stop re-learning the page structure every time they navigate and just focus on the content.

I also made two explicit design decisions in direct response to limitations in the synthetic dataset. When I discovered that the data had generated only one admission per hospital, I removed the hospital-level ranking visual entirely because the resulting chart would have been meaningless, and replaced it with insurance provider analysis which showed genuine variance. On the scatter plots on Pages 3 and 4 where the axis ranges are compressed, I added a small annotation to acknowledge that the narrow spread reflects the uniform nature of synthetic data rather than a real-world pattern. I made both of those calls because I think it is more important to be honest about what the data can and cannot support than to produce visuals that look impressive but mislead.

User Interaction

I designed the interaction model around two types of users: people who want to scan the dashboard for a quick read of the current state, and people who want to actively investigate a specific question. The structure supports both without forcing either type of user to change how they naturally work.

At the broadest level users move between pages using the tab navigation at the bottom. I named the pages Executive Summary, Detailed Analysis, Performance 1, and Performance 2 so that a user can immediately identify which page matches the question they are trying to answer. Each page is designed to be self-contained, meaning a user who navigates directly to Page 3 gets a complete picture of provider and payer performance without needing to have read Pages 1 and 2 first.

Within each page the slicers on the right panel are the main filtering mechanism. I gave each page its own slicer set relevant to that page's analytical focus: Page 1 has Year, Gender, and Admission Type; Page 2 adds Medical Condition and Age Group; Page 3 has Insurance

Provider and Admission Type; Page 4 uses a Year range slider alongside Quarter and Medical Condition. I set all slicers to default to All so that a first-time viewer sees the complete dataset before any filtering is applied. Changing any slicer immediately cross-filters every visual on the page at the same time, which means a user who spots something unusual on one chart can narrow to a specific condition or year and instantly see whether that pattern holds across all the other visuals on the page without having to navigate anywhere.

Cross-filtering is the most powerful exploration tool in the report and I wanted to make sure it worked intuitively. Clicking any element on any chart: a bar, a line point, a scatter bubble, a heatmap cell, filters every other visual on the page to that selection. For example, clicking Cancer on the medical condition bar on Page 1 immediately narrows the KPI cards, the trend line, the gender split, and the admission type chart to cancer patients only. Clicking the same element again releases the filter. This turns every chart into a filter and lets users follow an analytical thread visually rather than having to repeatedly adjust slicers.

On Page 4 I added a Year range slider specifically because the financial performance questions on that page often involve comparing windows of time rather than selecting individual years from a list. A user can drag either endpoint of the slider to isolate a specific period, which complements the categorical Year slicer used on other pages.

The final interaction layer is tooltips. I enabled tooltips on every visual so that hovering over any data point shows the exact underlying values. I made this decision because it keeps the charts themselves visually clean — data labels on every bar or point would create clutter — while still giving users access to precise numbers whenever they need them. This is especially important on the heatmap on Page 2 where the colour intensity shows the broad pattern but the exact admission counts matter for any follow-up analysis.

Part F: Analytical insights and business recommendations

Analytical Insights

1. Admissions growth is outpacing billing growth, which tells me there is margin compression happening per patient

When I looked at the Executive Summary the first pattern that stood out to me was that total admissions had grown 7.48% year on year, reaching 55,000 patients and generating $1.40bn in total billing. But when I looked at average billing per admission it was sitting at $25,590 and had stayed essentially flat across the four-year period visible in my billing by admission type and year chart. That divergence between volume growth and per-patient revenue growth is the single most important financial pattern I found in the dataset. What it tells me is that the hospital is serving more patients without proportionally increasing revenue per encounter, and if the cost of serving each patient, staffing, consumables, facility time which is rising with inflation then that creates real margin pressure. In a real hospital setting this pattern usually points to either a shift toward lower-reimbursement payer contracts or a shift in case mix toward fewer complex admissions.

2. The patient population is dominated by a single billing tier, which creates revenue concentration risk

When I built the patient distribution by billing category donut on Page 1, I found that 91.98% of admissions fall into the High billing category with only 8.02% in Low. My first reaction was that this looked positive meaning most patients generating high billing. But when I thought about it more carefully, I realised it actually signals a concentration risk. A hospital whose revenue depends almost entirely on one patient tier is exposed to any policy change, reimbursement adjustment, or demographic shift that affects that tier. It also made me question whether the billing categorisation itself is too coarse to be useful hence need a more granular split into quartiles or deciles would likely reveal meaningful sub-segments within the High category that could inform clinical pathway design.

3. Monthly admission volumes are volatile but show no discernible seasonal pattern

Looking at my monthly admission trends line chart I could see admissions fluctuating between roughly 1,350 and 1,650 per month across Elective, Emergency, and Urgent

categories. What struck me was that the three lines track each other closely month to month and there is no sustained peak in any particular quarter. I think the absence of seasonality is itself an operationally significant finding. Many hospitals plan staffing around predictable winter respiratory peaks or summer trauma increases but this hospital's admission pattern would not support that model. What the data is actually telling me is that capacity planning here needs to be built around rolling averages and demand variability rather than seasonal forecasts. The sharp dip in February Emergency admissions is also something I would want to investigate further on real data for it could reflect a staffing gap, a referral pipeline disruption, or just a reporting anomaly.

4. High risk patients consume disproportionate resources relative to their share of the population

On Page 4 my risk segment scatter plot shows a clear stratification that High Risk patients sit at the top of the length of stay axis at around 15.6 days average while Low and Medium Risk patients sit lower at around 15.45 days. Even within the narrow variance of the synthetic data the ordering is clinically coherent leading to higher risk associates with longer stay and moderately higher billing per admission. What this pattern is pointing toward, and what I would expect to see much more strongly on real data, is the well-documented 80/20 distribution of hospital resource consumption where a minority of high-acuity patients consume the majority of bed days and clinical time. For me this is where the largest cost-containment opportunity sits, identifying these patients earlier in their admission, or ideally before admission, is the intervention most likely to produce meaningful savings.

5. Payer mix looks balanced on the surface but admission type composition reveals differentiated utilisation patterns

When I looked at the admissions by insurance provider and admission type stacked bar on Page 3, I noticed that while all five major insurers carry roughly similar total admission volumes the internal mix of Elective, Emergency, and Urgent admissions is visibly different across providers. That differentiation matters to me because emergency admissions are typically more expensive and less predictable than elective ones. Insurers whose members are presenting disproportionately through emergency pathways may have gaps in their outpatient access, prior-authorisation processes, or chronic disease management programmes. I think this creates a basis for having insurer-specific conversations about care coordination rather than treating all payer relationships as identical.

6. Clinical outcomes do not strongly differentiate length of stay across conditions

The length of stay by medical condition and test results chart on Page 2 showed me average stays clustered tightly between roughly 15 and 16 days regardless of whether test results were Abnormal, Inconclusive, or Normal and regardless of condition. In a real clinical setting, I would find that concerning meaning abnormal test results should typically correlate with longer or more complex care episodes. The absence of that correlation here either reflects the synthetic nature of the data or, on real data, would indicate that test result severity is not being used effectively to stratify care pathways. Either way I think it is worth flagging to clinical governance.

Actionable Recommendations

1. Launch a per-admission revenue diagnostic programme

Given the gap between admission volume growth and flat per-patient billing I would recommend the hospital commission a three-part diagnostic in the next quarter. First, decompose per-admission billing by payer, condition, and admission type to identify which combinations are pulling the average down. Second, benchmark reimbursement rates against regional or national averages for the top five condition groups. Third, review the clinical coding and documentation process because under-coding of comorbidities is one of the most common reasons per-admission billing underperforms peers. The expected output is a prioritised list of three to five revenue improvement opportunities with quantified impact. In my view this is the highest value single intervention the dashboard points toward.

2. Build an insurer-level case management programme targeting high emergency-admission payers

The differentiated admission type mix I identified on Page 3 should translate into differentiated insurer engagement. For insurers whose members are disproportionately presenting through emergency rather than elective pathways I would recommend the hospital propose joint care coordination initiatives to have shared chronic disease registries, escalation protocols for high-risk members, and prior-authorisation streamlining for commonly delayed procedures. Emergency admissions that could have been managed electively represent both a clinical quality issue and a cost of care issue. Reducing even 5% of avoidable emergency presentations across the top two insurers would materially reduce capacity pressure. It also

positions the hospital as a proactive clinical partner which is increasingly important in value-based contracting environments.

3. Implement early risk stratification at admission to target the high-risk segment

Based on what I found in the risk segment analysis on Page 4, I would recommend developing a validated risk scoring protocol applied at the point of admission, or ideally at the point of referral, to flag High Risk patients for intensified case management. That means earlier multidisciplinary ward rounds, dedicated discharge planning from day one, and post-discharge follow-up within 72 hours. On real data, early intervention programmes of this type typically reduce length of stay by 0.5 to 1.5 days for high-risk patients and reduce 30-day readmission rates by 10 to 20%. Even conservative estimates would produce material bed-day savings given the admission volumes I observed.

4. Move from seasonal forecasting to a rolling variability-based planning model

Because my admission data shows no clear seasonality but significant month-to-month volatility I would recommend the hospital move away from quarterly seasonal forecasting and toward a rolling twelve-week demand model with explicit variance bands. In practice this means scheduling core permanent staffing for the rolling average and building a flexible pool of agency, overtime, and cross-trained staff sized to cover the variance rather than the peak. That is a different operational philosophy from what most hospitals use but it is the one that actually matches the pattern this data shows.

5. Review billing categorisation granularity

The 92/8 split between High and Low billing categories that I found is too coarse to drive meaningful analysis or segmentation. I would recommend clinical finance and analytics teams redefine billing categories into at least four tiers, for example, quartiles of billing amount and rebuild the associated reporting. This is a relatively low-cost data infrastructure change but it unlocks significantly more precise analysis on every subsequent question asked of the data.

Limitations and Data Quality Notes

I want to be transparent about the limitations of this analysis. It is based on a synthetic dataset and several categorical variables like medication, blood type, insurance provider, and hospital that appear to have been generated with uniform random assignment. As a result, a number of patterns I would expect to see on real hospital data do not appear here like payer concentration, seasonal admission peaks, condition-specific cost variance, and blood type clinical correlations are all flat in this dataset. The insights I have drawn are grounded in what the dashboard genuinely shows rather than what I would expect real data to show. The recommendations are framed to apply the analytical approach to real operational data where the underlying patterns would be considerably more pronounced. If this same dashboard architecture were applied to production data, I would expect the payer mix, seasonality, and risk stratification analyses in particular to surface sharper and more directly actionable findings than what is visible here.